

Point-by-Point Response to Reviewers

Manuscript: mjdrydypu_451_26 — “Can Ageing Be Slowed or Reversed? A Reproducible Evidence Map and Credibility Ranking of Anti-Ageing and Age-Reversal Interventions”

Journal: Medical Journal of Dr. D.Y. Patil Vidyapeeth (MJDRDYPUP)

Date: 23 June 2026

I thank the editor and the reviewer for their careful reading. Responses are given in table form below. Section and page references point to the revised manuscript.

A preliminary clarification. This manuscript is a reproducible bibliographic evidence map of anti-ageing intervention classes. It does not enrol patients, does not define diabetic or non-diabetic groups, does not administer cognitive instruments such as the MMSE, and holds no individual-level medication or socioeconomic data. Several reviewer comments (Comments 1, 2, and 4) describe features of a patient-level comparative clinical cohort — diabetic versus non-diabetic groups, MMSE-based cognitive labelling, and individual polypharmacy assessment — that are not part of this study. I respectfully raise the possibility that these comments may have been intended for a different manuscript. Rather than set them aside, I have engaged the underlying geriatric principles each comment raises and shown how they are handled within the scope of an evidence map. I would of course welcome any further reviewer comments specific to this manuscript.

Reviewer 1

#	Reviewer comment	Applicability to this manuscript	Response and action taken
1	“The diabetic and non-diabetic groups are reasonably balanced; however, important confounding variables such as functional status, frailty level, medication appropriateness, socioeconomic dependency are not fully controlled.”	Partially applicable in principle only. This study has no diabetic/non-diabetic groups and no individual-level covariates to control; it is a synthesis of published studies, not a comparative cohort.	I have clarified the study design in the Abstract and Methods so the absence of patient groups is unambiguous. The principle of confounding control is nonetheless central to appraising the primary studies synthesised here, and this is addressed through the structured risk-of-bias appraisal (Table 3), which includes an explicit “Confounding” domain and flags the observational five-cohort study as carrying residual confounding (Some concern). The Strengths and Limitations section now states that source-study confounding constrains the strength of the mapped signals. (Methods; Table 3; Strengths and Limitations.)

#	Reviewer comment	Applicability to this manuscript	Response and action taken
2	“Almost everyone was found to have cognitive problems in both groups... (MMSE) may not suit people with low education and whether adjustments were made for literacy level... some normal ageing may have been wrongly labelled as disease.”	Not applicable as stated; the principle is noted. No MMSE was administered and no participant was labelled cognitively impaired in this study; there are no diabetic/non-diabetic groups and no cognitive prevalence figure.	I have made explicit that no cognitive instrument was administered in this work. Where cognition appears, it is an outcome reported by an included trial (the PROMOTe microbiome RCT). I have added a note that cognitive and other surrogate outcomes drawn from primary trials require careful interpretation of the measurement instrument used — including the known sensitivity of brief screens such as the MMSE to education and literacy — and that biomarker or screening-test movement should not be over-interpreted as disease reversal. This reinforces the manuscript’s core distinction between biomarker change and clinical rejuvenation. (Results, microbiome paragraph; Discussion.)
3	“Frailty is one of the most important concepts in geriatrics which can help to explain falls, weakness, dependency, hospitalisation risk; a note may be added on it.”	Directly applicable and welcome. Frailty is already a central construct in this map.	I have strengthened the treatment of frailty, which is the dominant healthspan outcome across the highest-ranked evidence: the multimodal exercise trial (frailty index), the exergame frailty/sarcopenia trial, and the five-cohort frailty–mortality analysis. The Discussion now states explicitly that frailty integrates falls, weakness, dependency, and hospitalisation risk, and that the strongest human signal in this map — exercise — acts principally through frailty reduction. (Results, exercise paragraph; Discussion.)

#	Reviewer comment	Applicability to this manuscript	Response and action taken
4	“It is correct that diabetics take more medicines. But we should also ask: are all these medicines necessary?... unnecessary medicines can cause more harm than benefit, but necessary medications may not be focused upon if ‘essential’ rather than desirable.”	Not applicable as stated; the principle is noted. This study has no medication or polypharmacy dataset and no diabetic cohort.	The principle of medication appropriateness — distinguishing essential from merely desirable therapy and weighing benefit against harm — is one I fully endorse, and it aligns with the manuscript’s central argument that mechanistic or biomarker plausibility does not establish clinical benefit. I have made this connection explicit for the pharmacologic intervention classes (rapamycin, metformin, senolytics, supplements): the Discussion now states that, as in rational geriatric prescribing, these agents should not be adopted as “anti-ageing” therapy until benefit, necessity, dose, and safety are demonstrated in confirmatory human trials. (Discussion, pharmacologic paragraph.)

Additional improvements made on the author’s initiative

Beyond the reviewer comments, I undertook a thorough self-audit and strengthened the manuscript as follows.

#	Area	Issue in submitted version	Change made
1	Title	Promised a “Mechanistic Synthesis” that was not delivered as a discrete analysis.	Title refined to “...A Reproducible Evidence Map and Credibility Ranking of Anti-Ageing and Age-Reversal Interventions.”
2	Abstract conclusion	Ended with internal working language (“Final submission requires manual full-text eligibility confirmation...”).	Replaced with a definitive scholarly conclusion stating the evidence hierarchy and the healthspan-versus-rejuvenation distinction.
3	Scientific framing	Repeated “pilot,” “capped run,” and “counts will change” language undermined the work.	Reframed throughout as a completed, deliberately scoped evidence map; honest limitations retained, self-undermining language removed.
4	Figure 2 (data integrity)	The figure plotted the <code>max_claim_score</code> column (0–14 scale) while its caption and Table 1 used the <code>credibility_score</code> (exercise 31.48); the figure ranking disagreed with the table.	Figure 2 regenerated from the <code>credibility_score</code> column so figure, caption, and Table 1 now agree exactly; data labels added.

#	Area	Issue in submitted version	Change made
5	All figures	No data labels; “extracted repo evidence” workspace jargon on axes; overlapping point labels.	Figures 1–4 regenerated with clear axis titles, data labels, tier colour legends, and de-overlapped annotations; the PRISMA flow now shows the full screening breakdown with connected arrows.
6	References	All authors were listed for every reference (up to ~19 authors).	Truncated to the first six authors followed by “et al.” per Vancouver style.
7	authorship Citation sequence	Refs 15–40 (Methods) were first cited before refs 11–14 (Results), breaking sequential order.	The Introduction now cites the full human-evidence set [3–14] before the collective [15–40] citation, restoring strict sequential numbering.
8	In-text citation style	Inconsistent rendering of citation numerals; consecutive runs written out in full.	All in-text citations use superscript bracketed numerals; runs of three or more consecutive references collapsed to ranges (e.g., [3–14], [2,4–6], [3,7–9,12–14]) per Vancouver style.
9	Reference details	References lacked volume, issue, and page/article numbers, and one had an incorrect year.	Volume/issue/article and DOIs added for all entries where verifiable (PubMed/Crossref); ref 6 year corrected from 2026 to 2025; ref 12 DOI independently verified as correct.
10	Figure 1 formatting	Text overflowed the box borders in the study-selection flow diagram.	Figure 1 redrawn with wider, auto-sized boxes so all text fits cleanly within borders, with connected directional arrows.
11	Display-item placement	Tables and figures were embedded mid-text.	For typesetting clarity, Tables 1–3 and the figure legends are now placed after the references, and the figures are supplied as a separate file (Figures 1–4), each cited at first mention in the text.
12	Risk-of-bias framing	Table 3 was labelled as non-final “prompts.”	Reframed as a structured risk-of-bias appraisal following RoB 2 / ROBINS-I signalling logic, based on available text.

#	Area	Issue in submitted version	Change made
13	Declarations / data availability	Referred to “project files,” “screening artifacts,” and duplicated the Acknowledgements section.	Cleaned to a standard data-availability statement with the public repository link; duplicate sections removed; a stray character after the Conclusion deleted.
14	Language and tone	Several informal or workflow-oriented phrasings unsuited to a scientific article.	Revised throughout for formal academic register, grammar, and clarity.

I believe these revisions materially strengthen the manuscript and address the scientific principles raised by the reviewer. I am happy to make any further changes the editor or reviewer considers necessary.

Sincerely,

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